SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY
An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade
Kuniamuthur, Coimbatore – 641008
Phone : (0422)-2678001 (7 Lines) | Email : info@skcet.ac.in | Website : www.skcet.ac.in

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(Accredited by NBA)

RECORD OF LABORATORY WORK

25CS202 / Application Development Practices

Name of the Student : Aswath S

Register No. : 25CS575

Year/Branch/Section : I / CSE / H

Semester : II

Academic Year : 2025-2026

SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY
An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade
Kuniamuthur, Coimbatore – 641008
Phone : (0422)-2678001 (7 Lines) | Email : info@skcet.ac.in | Website : www.skcet.ac.in

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(Accredited by NBA)

BONAFIDE CERTIFICATE

Name of the Student: . . . . . . Aswath S **. . . . . .**

Register No : . . . . . .25CS575**. . . . . .**

Year : . . I **. . Branch : . . .** CSE **. . . . Section : . .** H **. . Semester : . . .** II **. . . .**

Certified that this is the bonafide record of work done by the above student in the **25CS202 / Application Development Practices** during the academic year 2025 - 2026.

Faculty Incharge **HOD**

Submitted for the End Semester Examination held on **. . . . . .**

Internal Examiner **External Examiner**

SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY
An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade
Kuniamuthur, Coimbatore – 641008
Phone : (0422)-2678001 (7 Lines) | Email : info@skcet.ac.in | Website : www.skcet.ac.in

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(Accredited by NBA)

INDEX

S.NO	LIST OF EXPERIMENTS	PAGE. NO	MARKS	SIGNATURE
1	Create a responsive web page with a registration form using HTMLS, CSS3, and Bootstrap grid system.			
2	Enhance web elements with CSS: Shadows, Rounded Borders, and Multiple Backgrounds.			
3	Implement a CSS grid-based layout to create a responsive web page.			
4	Implement a JavaScript-based form validation for user input fields (email, password, etc.).			
5	Develop a dynamic webpage using JavaScript and DOM manipulation.			
6	Work with APIs and fetch data asynchronously using JavaScript.			
7	Create a TypeScript application demonstrating object-oriented programming concepts.			
8	Write a Linux shell script to automate file management tasks.			
9	Schedule background jobs using Linux cron and batch scripting.			
10	Perform Git version control operations: cloning, committing, branching, and merging.			
11	Collaborate on a project using GitHub: Forking, pull requests, and resolving merge conflicts.			
12	Deploy a static website using GitHub Pages.			

SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY
An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade
Kuniamuthur, Coimbatore – 641008
Phone : (0422)-2678001 (7 Lines) | Email : info@skcet.ac.in | Website : www.skcet.ac.in

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(Accredited by NBA)

CONTINUOUS EVALUATION SHEET REFERENCES

RUBRICS TABLE

Criteria	Range of Marks			
	Excellent	**Good**	**Average**	**Below Average**
Aim & Algorithm (20)	18-20	17-18	15-16	0-14
Program (30)	27-30	21-26	15-20	0-14
Execution and Result (30)	27-30	21-26	15-20	0-14
Documentation (10)	9-10	7-8	5-6	0-4
Viva (10)	9-10	7-8	5-6	0-4
Overall Marks (100)	**90-100**	**70-89**	**50-69**	**0-49**

SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY
An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade
Kuniamuthur, Coimbatore – 641008
Phone : (0422)-2678001 (7 Lines) | Email : info@skcet.ac.in | Website : www.skcet.ac.in

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(Accredited by NBA)

RUBRICS BASED ASSESSMENT SHEET

Name of the Lab:

Reg. No.:

Name of the Student:

Components	EX1	EX2	EX3	EX4	EX5	EX6	EX7	EX8	EX9	EX10	EX11	EX12
Aim& Algorithm (20 Marks)												
Program (30 Marks)												
Execution and Result (30 Marks)												
Documentation (10 Marks)												
Viva (10 Marks)												
TOTAL												
Consolidated Mark (100)												
Faculty Signature												

EXP. NO: 1	Create a responsive web page with a registration form using HTMLS, CSS3, and Bootstrap grid system.
DATE:	

AIM:

To create a responsive web page with a registration form using HTMLS, CSS3, and Bootstrap grid system.

ALGORITHM:

1. **Page Setup**
    - Load HTML document with and language set to English.
    - Import Bootstrap CSS (for responsive design and styling).
    - Import custom stylesheet IPL.css.
2. **Body Layout**
    - Apply d-flex align-items-center py-4 to center content vertically with padding.
    - Create a container to hold the registration card.
3. **Card Creation**
    - Inside the container, a card (ipl-card) is created with padding and max-width.
    - The card has a header (ipl-header) containing:
    - A centered title: *"IPL Player Registration"*.
    - An IPL logo image (download.jpg).
4. **Form Structure**
    - A Bootstrap grid (row g-4) organizes form fields into columns.
    - Each input field has a label (form-label fw-bold) and a corresponding input element.
5. **Form Fields**
    - **Personal Info:**
        - First Name (text input, required).
        - Last Name (text input, required).
        - Email (email input, required).
        - Age (number input, required).
        - Date of Birth (date input, required).

        - Contact Number (tel input, required).
        - Nationality (text input, required).
    - **Player Details:**
        - Player Role (dropdown: Batsman, Bowler, etc.).
        - Bowling Style (dropdown: fast, spin, etc.).
        - Batting Style (dropdown: right/left/ambidextrous).
        - Experience (radio buttons: 7).
        - Previous IPL Teams (dropdown with team names).
    - **Confirmation:**
    - Checkbox to agree to terms & code of conduct (required).

6. **Form Submission**
    - A Submit button styled with Bootstrap + custom CSS.
    - On hover, button changes color and slightly enlarges (transform: scale(1.05)).
7. **CSS Styling**

- **Body:** Beige background.
- **Card (ipl-card):** Cyan background, rounded corners.
- **Header (ipl-header):** Navy background, yellow text, rounded corners.
- **Labels:** Styled with italic + bold.
- **Button:** Blue background, white text, rounded corners, hover effects

SOURCE CODE:

1) HTML Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>IPL Player Registration with BS</title>

  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">

  <link href="IPL.css" rel="stylesheet">
```

```
</head>
<body class="d-flex align-items-center py-4">
  <div class="container">
    <div class="card ipl-card p-4 p-xl-5 mx-auto" style="max-width: 960px;">

      <div class="ipl-header d-flex align-items-center justify-content-center">
        <h2 class="mb-2 text-center flex-grow-1">IPL Player Registration</h2>
        <img src="download.jpg" class="img-fluid" width="100" alt="IPL Logo">
      </div>

      <form class="row g-4 p-3">
        <div class="col-md-6">
          <label for="fullName" class="form-label fw-bold">First Name <span class="text-danger">*</span></label>
          <input type="text" class="form-control" id="fullName" placeholder="Enter your First Name" required>
        </div>

        <div class="col-md-6">
          <label for="fullName" class="form-label fw-bold">Last Name <span class="text-danger">*</span></label>
          <input type="text" class="form-control" id="fullName" placeholder="Enter your Last Name" required>
        </div>

        <div class="col-md-6">
          <label for="email" class="form-label fw-bold">Email<span class="text-danger">*</span></label>
          <input type="email" class="form-control" placeholder="Enter your Email" id="email" required>
        </div>

        <div class="col-md-3">
```

```
        <label for="age" class="form-label fw-bold">Age <span class="text-danger">*</span></label>

        <input type="number" class="form-control" id="age" placeholder="Enter your Age" maxlength="2" required>

      </div>


      <div class="col-md-3">

        <label for="dob" class="form-label fw-bold">Date of Birth <span class="text-danger">*</span></label>

        <input type="date" class="form-control" required>

      </div>


      <div class="col-md-6">

        <label for="phone" class="form-label fw-bold">Contact Number <span class="text-danger">*</span></label>

        <input type="tel" class="form-control" id="phone" maxlength="10" placeholder="Enter your Contact Number" required>

      </div>


      <div class="col-md-6">

        <label for="nation" class="form-label fw-bold">Nationality <span class="text-danger">*</span></label>

        <input type="text" class="form-control" id="nation" placeholder="Enter your Nationality" required>

      </div>


      <div class="col-md-12">

          <label for="playerRole" class="form-label fst-italic fw-bold">Player Role</label>

          <select class="form-select" id="role" required>

              <option>-- Select your Role --</option>

              <option>Batsman</option>

              <option>Bowler</option>

              <option>All-Rounder</option>
```

```
      <option>Wicket-Keeper</option>
    </select>
  </div>

  <div class="col-md-6">
    <label for="bowlhand" class="form-label fst-italic fw-bold">Bowling Style</label>
    <select class="form-select" id="bowlstyle" required>
      <option>-- Select your style --</option>
      <option>N/A (Pure Batter)</option>
      <option>Right-Handed Fast</option>
      <option>Right-Handed Medium</option>
      <option>Right-Handed Leg-Spin</option>
      <option>Right-Handed Off-Spin</option>
      <option>Left-Handed Fast</option>
      <option>Left-Handed Medium</option>
      <option>Left-Handed Leg-Spin</option>
      <option>Left-Handed Off-Spin</option>
    </select>
  </div>

  <div class="col-md-6">
    <label for="bathand" class="form-label fst-italic fw-bold">Batting Style</label>
    <select class="form-select" id="batstyle" required>
      <option>-- Select your style --</option>
      <option>Right-Handed</option>
      <option>Left-Handed</option>
      <option>Ambidextrous</option>
    </select>
  </div>

  <div class="col-md-12">
```

```
    <label for="experience" class="form-label fst-italic fw-bold">Experience</label>
    <div class="d-flex gap-5 pt-2">
      <div class="form-check">
        <input class="form-check-input" type="radio" name="experience">
        <label class="form-check-label"> Less than 3 Years</label>
      </div>
      <div class="form-check">
        <input class="form-check-input" type="radio" name="experience">
        <label class="form-check-label"> 3 to 5 years</label>
      </div>
      <div class="form-check">
        <input class="form-check-input" type="radio" name="experience">
        <label class="form-check-label"> 5-7 years</label>
      </div>
      <div class="form-check">
        <input class="form-check-input" type="radio" name="experience">
        <label class="form-check-label"> More than 7 years</label>
      </div>
    </div>
  </div>

  <div class="col-md-12">
    <label for="team" class="form-label fst-italic fw-bold">Previous IPL Teams (if any)</label>
    <select class="form-select" id="team" required>
      <option>-- Select your Team --</option>
      <option>N/A</option>
      <option>Chennai Super Kings</option>
      <option>Mumbai Indians</option>
      <option>Delhi Capitals</option>
      <option>Gujarat Giants</option>
      <option>Lucknow Super Giants</option>
```

```
            <option>Sunrisers Hyderabad</option>
            <option>Royal Challengers Bengaluru</option>
            <option>Rajastan Royals</option>
            <option>Kolkata Knight Riders</option>
            <option>Punjab Kings</option>
          </select>
        </div>

        <div class="col-12 mt-4">
          <div class="form-check">
            <input class="form-check-input" type="checkbox" id="termsCheck" required>
            <label class="form-check-label fst-italic fw-semibold" for="termsCheck">
              I confirm that the information provided is accurate and I agree to the IPL player terms
& code of conduct. <span class="text-danger">*</span>
            </label>
            <div class="invalid-feedback">You must accept to register.</div>
          </div>
        </div>

        <div class="col-12">
          <button type="submit" class="col-12 d-flex align-items-center justify-content-
center">Submit</button>
        </div>
      </form>
    </div>
  </div>
</body>
</html>
```

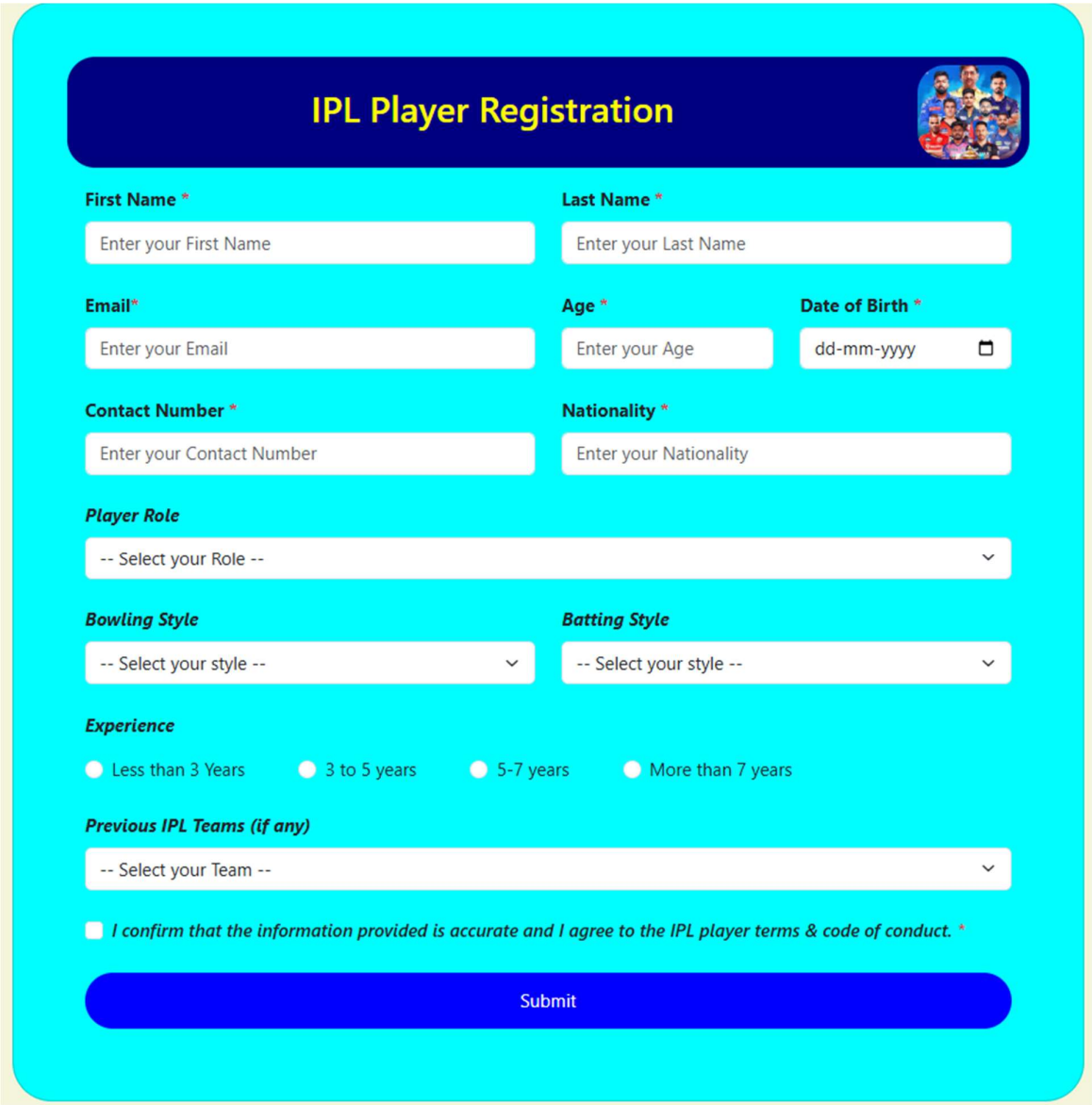
2) CSS Code:

```
body{
   background-color: beige;
```

```
}

.ipl-card{
   background-color: cyan;
   border-radius: 35px;
   border-color: 10px solid white;
}

.ipl-header{
   background: navy;
   border-radius: 25px;
   height: 100px;

   h2{
      text-align: center;
      color: yellow;
   }

   img{
      border-radius: 25px;
      padding-right: 7px;
   }
}

.form-label i {
         color: #1e3c72;
         width: 22px;
         margin-right: 6px;
      }

button{
```

```
    border: none;
    background-color: blue;
    color: white;
    border-radius: 25px;
    height: 50px;
    transition: background-color 0.3s ease,
            color 0.3s ease,
            transform 0.3s ease;
}

button:hover{
    transition: 0.2s ease;
    background-color: white;
    color: black;
    transform: scale(1.05);
}
```

OUTPUT:



RESULT:

Hence the output for the sample code using HTML, CSS and Bootstrap has been displayed.

EXP. NO: 2	Enhance web elements with CSS: Shadows, Rounded Borders, and Multiple Backgrounds.
DATE:	

AIM:

To enhance web elements with CSS: Shadows, Rounded Borders, and Multiple Backgrounds.

ALGORITHM:

1. Page Initialization
    - Load HTML document with and set language to English.
    - Define metadata (charset, viewport) and page title ("Shadow").
    - Apply internal CSS styles.
2. Body Layout
    - Use flexbox (display: flex; align-items: center; justify-content: center;) to center content both vertically and horizontally.
    - Set body height to full viewport (100vh).
3. Profile Card (.shadow)
    - Create a div with class shadow.
    - Apply styles:
    - Dotted border with rounded corners (border-radius: 50px).
    - Different border colors (top = blue, bottom = red).
    - Box shadow (3px 3px 30px blue) for glowing effect.
    - Flexbox column layout for stacking elements vertically.
    - Bold, large font with a modern typeface.
4. Card Content
    - Name Label (.pa1)
        - Displays "Aswath S".
        - Styled with red text shadow (3px 3px 3px rgb(255,0,0)).
    - Profile Image
        - Displays "My Photo.jpg".
        - Image height = 200px, rounded corners (border-radius: 50px).
    - About Me Label (.pa2)

        - Displays "About Me..".
        - (Note: .pa2 has no CSS defined, so it inherits default styles).
    - Paragraph (.p1)
    - Contains personal details: location, date/time, favorites, hobbies, personality.
    - Styled with medium font size and centered text.

5. User Interaction

- Page is static (no JavaScript).
- User simply views the profile card with styled text, image, and shadow effects

SOURCE CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Shadow</title>
  <style>
    body{
      display: flex;
      align-items: center;
      justify-content: center;
      height: 100vh;
    }
    .shadow{
      display: flex;
      flex-direction: column;
      align-items: center;
      justify-content: center;
      border-style: dotted;
      height: auto;
      width: 600px;
```

```
        font-weight: bolder;
        font-size: xx-large;
        font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
        border-radius: 50px;
        border-top-color: blue;
        border-bottom-color: red;
        box-shadow: 3px 3px 30px blue;
      }

      .pa1{
        text-shadow: 3px 3px 3px rgb(255, 0, 0);
      }

      .p1{
        font-size: medium;
        text-align: center;
      }

      img{
        height: 200px;
        border-radius: 50px;
      }
    </style>
  </head>
  <body>
    <div class="shadow">
      <label class="pa1">Aswath S</label>
      <img src="My Photo.jpg"><br>
      <label class="pa2">About Me..</label>
      <p class="p1">
```

```
        Hello, I am from Coimbatore South, Tamil Nadu, India. <br>It's currently a pleasant Thursday afternoon, 12 March 2026.

        <br>- Favorite Food: Rasam With Potato Fry

        <br>- Favorite Car: Koenigsegg Agera RS

        <br>- Hobby: Biking

        <br>- Personality: I enjoy learning, exploring new ideas, and expressing myself in creative ways.

      </p>

   </div>

</body>

</html>
```

OUTPUT:

RESULT:

From the source code, the web elements are enhanced using CSS.

EXP. NO: 3	Implement a CSS grid-based layout to create a responsive web page.
DATE:	

AIM:

To implement a CSS grid-based layout to create a responsive web page.

ALGORITHM:

1. **Page Initialization**
    - Load HTML document with and set language to English.
    - Define metadata (charset, viewport) and page title ("CSS Grid").
    - Import Bootstrap CSS for responsive grid system.
    - Apply internal CSS for font family and size.
2. **Container Setup**
    - Create a container-fluid to allow full-width layout.
    - Inside it, define a row with padding (p-5).
3. **Row Structure**
    - The row is divided into multiple columns using Bootstrap's grid system.
4. **Header Section**
    - col-12: Full-width column.
    - Centered content (d-flex align-items-center justify-content-center).
    - Bordered with green (border border-3 border-success).
    - Displays label: **"Aswath S"**.
5. **Left Column**
    - col-md-4: Occupies 4/12 columns on medium+ screens.
    - Centered content with flexbox.
    - Contains an unordered list (ul) with items: **Hello, World, Universe**.
    - List style removed (list-style-type: none).
6. **Right Column**
    - col-md-8: Occupies 8/12 columns on medium+ screens.
    - Centered content with flexbox.

- Contains a paragraph (p) with personal details:
- Location, date/time.
- Favorite food, car, hobby.
- Personality description.

7. **Footer Section**

- col-12: Full-width column.
- Centered content with flexbox.
- Displays label: **“Aswath SKCET ©”**.
- Styled with green border.

SOURCE CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CSS Grid</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
  <style>
    body{
      font-family: 'Gill Sans', 'Gill Sans MT', Calibri, 'Trebuchet MS', sans-serif;
      font-size: 24px;
    }
  </style>
</head>
<body>
  <div class="container-fluid">
    <div class="row p-5">
      <div class="col-12 d-flex align-items-center justify-content-center border border-3 border-
success">
```

```
            <label>Aswath S</label>
          </div>
          <div class="col-md-4 d-flex align-items-center justify-content-center border border-3 border-success">
            <ul style="list-style-type: none;">
              <li>Hello</li>
              <li>World</li>
              <li>Universe</li>
            </ul>
          </div>
          <div class="col-md-8 d-flex align-items-center justify-content-center border border-3 border-success">
            <p>Hello, I am from Coimbatore South, Tamil Nadu, India. It’s currently a pleasant Thursday afternoon, 12 March 2026.
              <br>- Favorite Food: Rasam With Potato Fry
              <br>- Favorite Car: Koenigsegg Agera RS
              <br>- Hobby: Biking
              <br>- Personality: I enjoy learning, exploring new ideas, and expressing myself in creative ways.</p>
          </div>
          <div class="col-12 d-flex align-items-center justify-content-center border border-3 border-success">
            <label>Aswath || SKCET &copy;</label>
          </div>
        </div>
      </div>
</body>
</html>
```

OUTPUT:

Aswath S	
Hello World Universe	Hello, I am from Coimbatore South, Tamil Nadu, India. It's currently a pleasant Thursday afternoon, 12 March 2026. - Favorite Food: Rasam With Potato Fry - Favorite Car: Koenigsegg Agera RS - Hobby: Biking - Personality: I enjoy learning, exploring new ideas, and expressing myself in creative ways.
Aswath || SKCET ©	

RESULT:

Responsive web page is created by implementing CSS Grid-based layout.

EXP. NO: 4	Implement a JavaScript-based form validation for user input fields (email, password, etc.).
DATE:	

AIM:

To Implement a JavaScript-based form validation for user input fields (email, password, etc.).

ALGORITHM:

1. **Page Setup**
    - Load HTML document with <!DOCTYPE html>.
    - Define metadata (charset, viewport) and page title ("Form Validation").
    - Apply CSS styles to center the form and style the fieldset.
2. **Form Layout**
    - Create a fieldset with a legend titled "Form Validation".
    - Inside the form:
    - Input field for **Contact** (id="phone").
    - Input field for **Email** (id="email").
    - A **Validate** button that triggers the validate() function.
    - A element (id="res") to display the result.
3. **Validation Function (validate())**
    - Retrieve values from input fields:
        - ph ← phone number.
        - em ← email address.
    - Define regex patterns:
    - pat = ^[0-9]{10}$ → ensures phone has exactly 10 digits.
    - pat1 = ^[A-Za-z0-9]+@[a-z]+(\.[a-z]{2,})+$ → ensures email format is valid.
4. **Validation Logic**
    - If **phone OR email** does not match their respective patterns:
        - If phone is invalid:
            - Show alert: *"Enter the contact correctly!!"*.
            - Set result text = *"Invalid Contact"*.

        - Else (email invalid):
        - Show alert: *"Enter the email correctly!!"*.
        - Set result text = *"Invalid Email"*.
    - Else (both valid):
    - Show alert: *"Details Validated Successfully"*.
    - Set result text = *"Valid Details"*.

5. **Result Display**

- Update the element with the validation result.

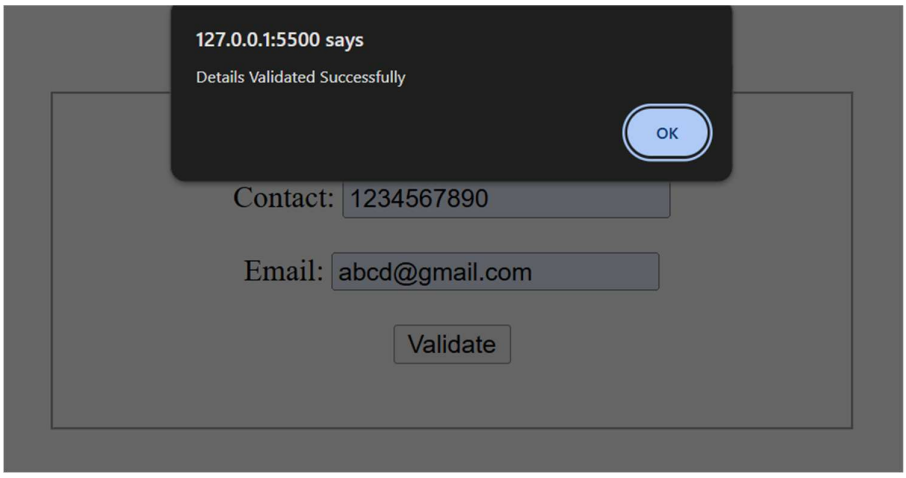
SOURCE CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Form Validation</title>
  <style>
    body{
      display: flex;
      align-items: center;
      justify-content: center;
    }

    form{
      text-align: center;
    }

    fieldset{
      height: 200px;
      width: 400px;
    }
```

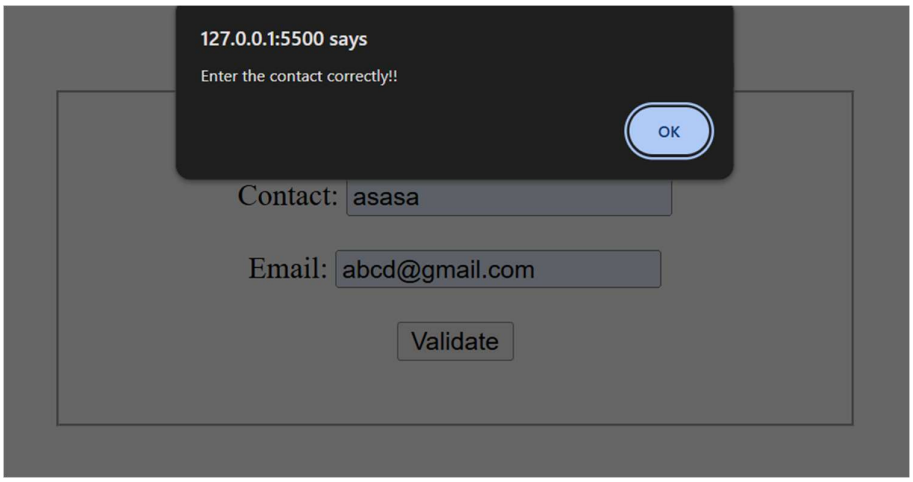
```
      legend{
        text-align: center;
      }
    </style>
    <script>
      function validate(){
        var ph=document.getElementById("phone").value;
        var em=document.getElementById("email").value;
        var pat=/^[0-9]{10}$/;
        var pat1=/^[A-Za-z0-9]+@[a-z]+(\.[a-z]{2,})+$/;
        if(!pat.test(ph) || !pat1.test(em)){
          if(!pat.test(ph)){
            alert("Enter the contact correctly!!");
            var res="Invalid Contact";
          }
          else{
            alert("Enter the email correctly!!");
            var res="Invalid Email";
          }
        }
        else{
          alert("Details Validated Successfully");
          var res="Valid Details";
        }
        document.getElementById("res").innerHTML=res;
      }
    </script>
  </head>
  <body>
    <fieldset>
```

```
    <legend><h1>Form Validation</h1></legend>
    <form>
      <label>Contact: </label><input type="text" id="phone"><br><br>
      <label>Email: </label><input type="text" id="email"><br><br>
      <button onclick="validate()">Validate</button>
      <p id="res"> </p>
    </form>
  </fieldset>
</body>
</html>
```

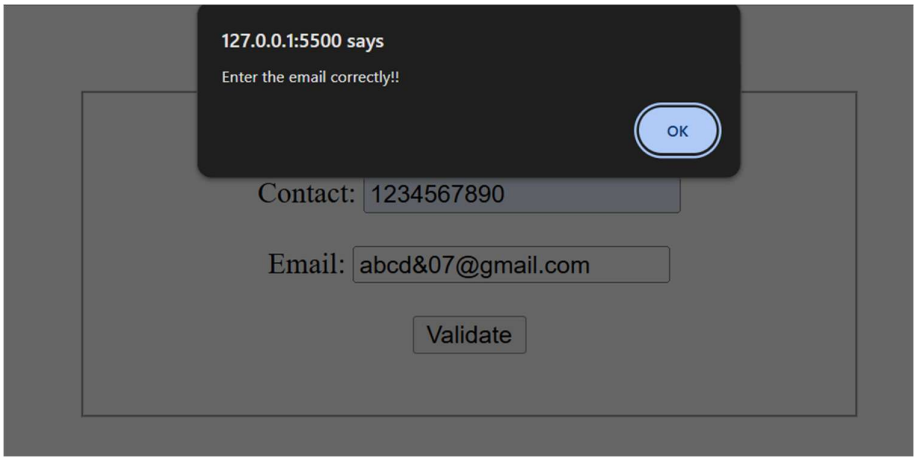
OUTPUT:



Form Validation
Contact: asasa
Email: abcd@gmail.com
Validate


127.0.0.1:5500 says
Enter the contact correctly!!
OK
Contact: asasa
Email: abcd@gmail.com
Validate

Form Validation
Contact: 1234567890
Email: abcd&07@gmail.com
Validate



RESULT:

Responsive web page is created by implementing CSS Grid-based layout.

EXP. NO: 5	Develop a dynamic webpage using JavaScript and DOM manipulation.
DATE:	

AIM:

To develop a dynamic webpage using JavaScript and DOM manipulation.

ALGORITHM:

1. **Page Setup**
   - Load HTML document with <!DOCTYPE html>.
   - Import external CSS file (Calculator.css).
   - Set page title to "Calculator".
2. **Heading**
   - Display a heading label: **"Calculator"** styled with large bold font.
3. **Input Section**
   - Two text input fields:
     - Num-1 (id="n1")
     - Num-2 (id="n2")
   - Both inputs allow user to enter numbers.
4. **Operation Buttons**
   - Four buttons are displayed:
   - **Add** → calls sum()
   - **Subtract** → calls sub()
   - **Multiply** → calls mul()
   - **Divide** → calls div()
5. **Result Display**
   - A label: **"Result:"**
   - Another label (id="res2") where the output will be shown.
6. **JavaScript Functions**
   - **sum()**
     - Get values from n1 and n2.
     - Convert to integers (parseInt).

        - Display result as "Sum : a+b".
    - **sub()**
        - Get values from n1 and n2.
        - Display result as "Difference : a-b".
    - **mul()**
        - Get values from n1 and n2.
        - Display result as "Product : a*b".
    - **div()**
    - Get values from n1 and n2.
    - If n2 != 0, display "Quotient : a/b".
    - Else, show alert "Division by Zero is not possible".

7. **CSS Styling**

- **Body:** Centered layout with red border.
- **Heading (.hh):** Large bold font.
- **Labels (.lab):** Margin spacing.
- **Input Section (.inp):** Flexbox centered with bold font.
- **Operation Buttons (#b1, #b2, #b3, #b4):**
    - Each button has unique colors and font size.
- **Result (.res, #res2):**
- Styled with padding and larger font.

SOURCE CODE:

1) HTML Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link rel="stylesheet" href="Calculator.css">
  <title>Calculator</title>
</head>
```

```
<body>
  <div class="heading">
    <label class="hh">Calculator</label>
  </div>
  <div class="inp">
    <label for="n1" class="lab">Num-1 : </label><input type="text" class="lab" id="n1"
size="30"><br><br>
    <label for="n2" class="lab">Num-2 : </label><input type="text" class="lab" id="n2"
size="30"><br><br>
  </div>
  <div class="operation">
    <input type="button" id="b1" value="Add" onclick="sum()">
    <input type="button" id="b2" value="Subtract" onclick="sub()">
    <input type="button" id="b3" value="Multiply" onclick="mul()">
    <input type="button" id="b4" value="Divide" onclick="div()">
  </div>

  <br><br><br>
  <label class="res">Result: </label><label id="res2"></label>
  <script>
    function sum(){
      let a=parseInt(document.getElementById("n1").value);
      let b=parseInt(document.getElementById("n2").value);
      document.getElementById("res2").innerText="Sum : "+(a+b);
    }

    function sub(){
      let a=parseInt(document.getElementById("n1").value);
      let b=parseInt(document.getElementById("n2").value);
      document.getElementById("res2").innerText="Difference : "+(a-b);
    }
```

```
    function mul(){
      let a=parseInt(document.getElementById("n1").value);
      let b=parseInt(document.getElementById("n2").value);
      document.getElementById("res2").innerText="Product : "+(a*b);
    }

    function div(){
      let a=parseInt(document.getElementById("n1").value);
      let b=parseInt(document.getElementById("n2").value);
      if(b!=0){
        document.getElementById("res2").innerText="Quotient : "+(a/b);
      }
      else{
        alert("Division by Zero is not possible");
      }
    }
  </script>
</body>
</html>
```

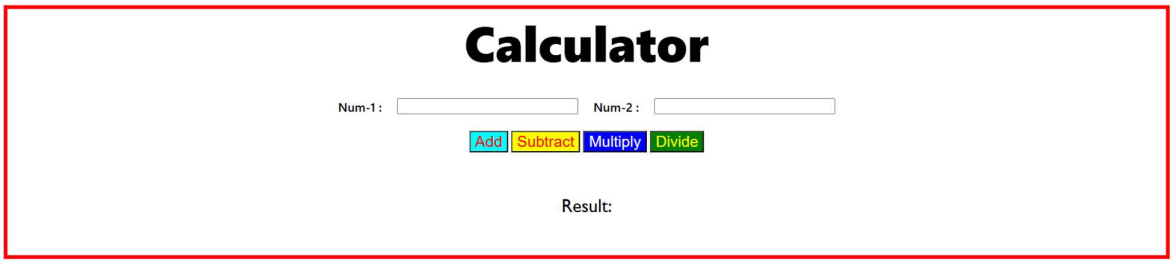
2) CSS Code:

```
body{
  display: flex;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  border: 5px solid red;
  margin-top: 100px;
}

.hh{
```

```
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    font-weight: 900;
    font-size: 64px;
}

.lab{
    margin: 10px;
}

.inp{
    display: flex;
    margin: 10px;
    align-items: center;
    justify-content: center;
    margin-top: 20px;
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    font-weight: 600;
}

.operation{
    border: none;
    #b1{
        color: red;
        background-color: aqua;
        font-size: 20px;
    }

    #b2{
        color: red;
        background-color: yellow;
        font-size: 20px;
```

```
    }

    #b3{
        color: white;
        background-color: blue;
        font-size: 20px;
    }

    #b4{
        color: yellow;
        background-color: green;
        font-size: 20px;
    }
}

.res{
    padding-bottom: 30px;
    font-size: x-large;
    font-family: 'Gill Sans', 'Gill Sans MT', Calibri, 'Trebuchet MS', sans-serif;
}

#res2{
    padding-bottom: 20px;
    font-size: 24px;
}
```

OUTPUT:



RESULT:

A Dynamic web page is created using JavaScript and DOM manipulation.

EXP. NO: 6	Work with APIs and fetch data asynchronously using JavaScript.
DATE:	

AIM:

To work with APIs and fetch data asynchronously using JavaScript.

ALGORITHM:

1. **Start the program**
   Load the HTML page in a browser.
2. **Display button interaction**
   Wait for the user to click the **DISPLAY** button.
3. **Trigger function**
   When the button is clicked, call the display () function.
4. **Fetch data from API**
   - Send an asynchronous request to the URL: https://jsonplaceholder.typicode.com/users
   - Wait for the response.
5. **Convert response to JSON**
   Parse the response into a JavaScript object (data).
6. **Initialize output string**
   Create an empty string variable output.
7. **Iterate through user data**
   For each user in the data array:
   - Extract the name field.
   - Extract the city field from address.city.
   - Append both values to the output string with formatting.
8. **Update HTML content**
   Insert the output string into the element with id "result".
9. **End program**
   The names and cities of all users are displayed on the webpage.

SOURCE CODE:

```
<!DOCTYPE html>
<html>
  <body>
    <H1> API INTERFACE PROGRAM</H1>
    <br><br>
    <button onclick="display()">DISPLAY</button>
    <br>
    <p id="result"></p>
    <script>
     async function display()
      {
        let info=await fetch("https://jsonplaceholder.typicode.com/users");
        let data=await info.json();
        let output="";
        for(let i=0;i<data.length;i++)
      {
            output+="<br>Name:"+ data[i].name+ "<br>City:"+ data[i].address.city+"<br>";
      }
        document.getElementById("result").innerHTML=output;
      }
    </script>
  </body>
</html>
```

OUTPUT:

API INTERFACE PROGRAM

DISPLAY

Name:Leanne Graham
City:Gwenborough

Name:Ervin Howell
City:Wisokyburgh

Name:Clementine Bauch
City:McKenziehaven

Name:Patricia Lebsack
City:South Elvis

Name:Chelsey Dietrich
City:Roscoeview

Name:Mrs. Dennis Schulist
City:South Christy

Name:Kurtis Weissnat
City:Howemouth

Name:Nicholas Runolfsdottir V
City:Aliyaview

Name:Glenna Reichert
City:Bartholomebury

Name:Clementina DuBuque
City:Lebsackbury

RESULT:

JavaScript enables smooth interaction with APIs through asynchronous data fetching. Using fetch() and promises, data can be retrieved and processed without blocking the UI. This approach enhances responsiveness and demonstrates real-time integration of external data into web applications.

EXP. NO: 7	Create a TypeScript application demonstrating object-oriented programming concepts.
DATE:	

AIM:

To create a TypeScript application demonstrating object-oriented programming concepts.

ALGORITHM:

1. **Define Person Class**
    1. Properties:
        - name → public
        - age → private
        - city → public
    2. Constructor initializes name, age, and city.
    3. Methods:
    4. setAge(a:number) → updates private age.
    5. getAge() → returns private age.
    6. display() → prints name, age (via getter), and city.
2. **Define Employee Class (extends Person)**
    1. Inherits all properties and methods of Person.
    2. Adds new properties:
        - empId → public
        - empSal → public
    3. Constructor:
        - Calls super(n,a,c) to initialize inherited properties.
        - Initializes empId and empSal.
    4. Overrides display():
    5. Calls super.display() to print Person details.
    6. Prints empId and empSal.
3. **Object Creation**
    1. Create an Employee object:

obj = new Employee("Aswath", 18, "Coimbatore", 1001, 15000);

This initializes:

2. name = "Aswath"
3. age = 18
4. city = "Coimbatore"
5. empId = 1001
6. empSal = 15000

4. **Method Execution**

- Call obj.display().
- Execution flow:
- Employee.display() runs.
- Calls Person.display() → prints name, age, city.
- Then prints employee ID and salary.

SOURCE CODE:

```
class Person{
  public name:string;
  private age:number;
  public city:string;

  constructor(n:string, a:number, c:string){
    this.name=n;
    this.age=a;
    this.city=c;
  }

  setAge(a:number):void{
    this.age=a;
  }
  getAge():number{
    return this.age;
```

```
  }

  display():void{
    console.log("Name:",this.name,"\nAge:",this.getAge(),"\nCity:",this.city);
  }
}

class Employee extends Person{
  public empId:number;
  public empSal:number;

  constructor(n:string, a:number, c:string, ei:number, es:number){
    super(n,a,c);
    this.empId=ei;
    this.empSal=es;
  }

  display():void{
    super.display();
    console.log("Employee ID:",this.empId,"\nSalary:",this.empSal);
  }
}

var obj=new Employee("Aswath",18,"Coimbatore",1001,15000);
obj.display();
```

OUTPUT:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

C:\Program Files\nodejs\node.exe --experimental-network-inspection .\TypeScript\OOPS.ts
Name: Aswath
Age: 18
City: Coimbatore
Employee ID: 1001
Salary: 15000
```

RESULT:

A TypeScript application is created demonstrating object-oriented programming concepts.

EXP. NO: 8	Write a Linux shell script to automate file management tasks.
DATE:	

AIM:

To write a Linux shell script to automate file management tasks.

ALGORITHM:

1. **Create a new directory**
   - Command: mkdir abc
   - Action: A folder named abc is created in the current working directory.
2. **Change into the new directory**
   - Command: cd abc
   - Action: The shell moves into the abc directory.
3. **Create a new text file and add content**
   - Command: cat > file1.txt
   - Action: Opens file1.txt for writing.
   - Input: You typed Aswath S and 25CS575.
   - Note: The process was stopped before saving all input, so only Aswath S was written.
4. **Display the file contents**
   - Command: cat file1.txt
   - Action: Shows the content of file1.txt → Aswath S.
5. **Change file permissions**
   - Command: chmod 777 file1.txt
   - Action: Grants read, write, and execute permissions to all users for file1.txt.
6. **List files with details**

- Command: ls -l
- Action: Displays file details (permissions, owner, size, date, filename).
- Output: Shows file1.txt with permissions -rw-r--r-- (indicating the permission change didn't persist as expected in your environment).

SOURCE CODE:

```
TUFFY@NightRaptor MINGW64 ~
$ mkdir abc

TUFFY@NightRaptor MINGW64 ~
$ cd abc

TUFFY@NightRaptor MINGW64 ~/abc
$ cat > file1.txt
Aswath S
25CS575
[2]+  Stopped                 cat > file1.txt

TUFFY@NightRaptor MINGW64 ~/abc
$ cat file1.txt
Aswath S

TUFFY@NightRaptor MINGW64 ~/abc
$ chmod 777 file1.txt

TUFFY@NightRaptor MINGW64 ~/abc
$ ls -l
total 1
-rw-r--r-- 1 TUFFY 197121 9 Mar 29 13:24 file1.txt
```

OUTPUT:

```
TUFFY@NightRaptor MINGW64 ~/ABCD
$ chmod 777 abcd.txt

TUFFY@NightRaptor MINGW64 ~/ABCD
$ ls -l
total 1
-rw-r--r-- 1 TUFFY 197121 30 Mar 29 13:31 abcd.txt
```

RESULT:

This experiment shows that Linux shell commands can efficiently handle directory creation, file editing, permission changes, and file listing.
It confirms that proper execution of commands ensures successful completion of basic file management tasks.